Feedback Lunch: Deep Feedback Codes for Wiretap Channels

Yingyao Zhou, *Student Member, IEEE*, Natasha Devroye, *Fellow, IEEE*, and Onur Günlü, *Senior Member, IEEE*

***Abstract*—We consider reversely-degraded wiretap channels, for which the secrecy capacity is zero if there is no channel feedback. This work focuses on a seeded modular code design for the Gaussian wiretap channel with channel output feedback, combining universal hash functions for security and learned feedback-based codes for reliability to achieve positive secrecy rates. We study the trade-off between communication reliability and information leakage, illustrating that feedback enables agreeing on a secret key shared between legitimate parties, overcoming the security advantage of the wiretapper. Our findings also motivate code designs for sensing-assisted secure communication, to be used in next-generation integrated sensing and communication methods.**

***Index Terms*—Wiretap channel with feedback, modular coding, reversely degraded channels, Gaussian wiretap.**

I. Introduction

SECURE communication can be achieved through key-based cryptography, which relies on securely-shared secret keys, and physical-layer security (PLS), which leverages the physical properties of the channel to ensure data confidentiality [1], [2]. Wyner introduced the wiretap channel (WTC), a fundamental model in PLS that enables reliable and secure communication when the eavesdropper's channel is a degraded version of the legitimate receiver's channel, resulting in a positive secrecy capacity [3]. Although feedback does not increase the channel capacity of memoryless channels, it improves the secrecy capacity [4], [5]. Here, we focus on the most challenging wiretap channel case, for which we design neural feedback codes. In this case, the eavesdropper has a statistical advantage over the legitimate receiver, i.e., reversely-degraded wiretap channel with output feedback (RD-WTC-F).

The modular coding scheme in [6] treats the security and reliability separately. Several works [7], [8] build on this idea for the Gaussian WTC without feedback, whereas feedback is a fundamental component in, e.g., mono-static integrated sensing and communication (ISAC) applications [5], [9]. Recently, deep-learned error-correcting feedback codes (DL-ECFCs) have been developed, such as in [10], for the additive white Gaussian noise channel with output feedback (AWGN-F), outperforming analytical linear codes in most cases.

Main Contributions: We extend the modular coding design [6] by combining universal hash functions with the current state-of-the-art DL-ECFC, Lightcode [10], to consider Gaussian WTC-F. We design and compare multiple feedback codes for the challenging case where the eavesdropper experiences the same or lower noise variance than the legitimate receiver (i.e. reversely-degraded setting). Non-feedback codes fail to achieve a positive secrecy rate in this setting. In contrast, our feedback coding scheme achieves positive secrecy rates with high reliability by leveraging feedback to generate secret keys shared between legitimate parties. Even with noisy feedback, legitimate parties achieve a security advantage gain over the eavesdropper. We refer to these feedback gains as a "feedback lunch", akin to a somewhat "free lunch" as opposed to the no-free-lunch theorems in security applications [11]. Moreover, to further enhance the secrecy performance, we introduce a new loss function incorporating an information-leakage constraint, enabling the learned model to actively reduce leakage while maintaining high reliability.

Notation: Random variables are denoted by capital letters, specific realizations by lowercase letters, and vectors in boldface. $\mathbb{F}_2$ is the binary field $\{0,1\}$, $\mathbb{R}^n$ the n-dimensional real vectors, and $\mathbb{R}_+$ the set of positive reals. $[a:b:c]$ denotes $[a, a+b, a+2b, \dots, c]$ with increment b and $a<c$. Subscripts denote time indices, and superscripts indicate the length of a vector, e.g., $\mathbf{Y}^n = (Y_1, \dots, Y_n)$. We use $\mathrm{card}(\mathcal{X})$ to denote the cardinality of the set $\mathcal{X}$ and define $[x]^+ = \max\{x, 0\}$. $Q(x)$ is the complementary distribution function of $\mathcal{N}(0,1)$, i.e., $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty \exp\left(-\frac{u^2}{2}\right) du$.

II. System Model

Consider a real-valued Gaussian RD-WTC-F with $\mathbb{P}(Y,Z|X)$, as depicted in Fig. 1. The transmitter, Alice, sends a message $\mathbf{M} \in \mathbb{F}_2^k$ to the legitimate receiver, Bob, over n channel uses, observed also by the eavesdropper, Eve. The code rate is defined as $R = k/n$. At each channel use $i \in [1:1:n]$, the forward channels to Bob and Eve are given by

$$Y_i = X_i + N_{Y,i}, \qquad Z_i = X_i + N_{Z,i} \tag{1}$$

where $N_{Y,i} \sim \mathcal{N}(0, \sigma_{Y,f}^2)$ and $N_{Z,i} \sim \mathcal{N}(0, \sigma_{Z,f}^2)$ denote independent and identically distributed (i.i.d.) Gaussian noise components. The transmitted symbols $X_i \in \mathbb{R}$ are subject to the average power constraint $\frac{1}{n}\mathbb{E}\left(\sum_{i=1}^n X_i^2\right) \le P$. We assume that there is channel output feedback available at the transmitter, which is delayed and possibly noisy from Bob and Eve. Such feedback could be obtained via reflections from both Bob and Eve, which does not necessarily require an active transmission from them [9]. Feedback can be either noiseless or noisy, with the noisy case modeled as $\tilde{Y}_{i-1} = Y_{i-1} + \tilde{N}_{Y,i-1}$ and $\tilde{Z}_{i-1} = Z_{i-1} + \tilde{N}_{Z,i-1}$, where $\tilde{N}_{Y,i-1} \sim \mathcal{N}(0, \sigma_{Y,fb}^2)$ and $\tilde{N}_{Z,i-1} \sim \mathcal{N}(0, \sigma_{Z,fb}^2)$. Define the signal-to-noise ratios

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Y. Zhou and N. Devroye are with the University of Illinois Chicago, Chicago, IL 60607, USA (Email: {yzhou238, devroye}@uic.edu)

O. Günlü is with the Lehrstuhl für Nachrichtentechnik, TU Dortmund, 44227 Dortmund, Germany and Information Coding Division, Linköping University, 581 83 Linköping, Sweden (E-mail: onur.guenlue@tu-dortmund.de).

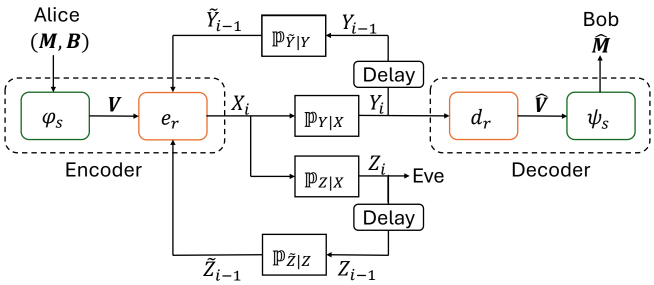

Fig. 1. Design of modular deep-learned feedback wiretap codes. The security and reliability layers are given by (φ_s, ψ_s) and (e_r, d_r), respectively.

(SNRs) for Bob as $\mathrm{S}_{Y,f} = P/\sigma^2_{Y,f}$ and $\mathrm{S}_{Y,fb} = P/\sigma^2_{Y,fb}$, and similarly for Eve as $\mathrm{S}_{Z,f}$ and $\mathrm{S}_{Z,fb}$, respectively.

As shown in Fig. 1, we consider the modular coding scheme proposed in [6], which consists of a security layer (φ_s, ψ_s) and a reliability layer (e_r, d_r). Separating these two layers facilitates independent control of information leakage and reliability. The security layer aims to control the leakage of the message $\mathbf{M}$ to Eve through her observations $\mathbf{Z}^n$. The security encoder uses a uniformly distributed random bit sequence $\mathbf{B} \in \mathbb{F}_2^{q-k}$ to produce $\mathbf{V} = \varphi_s(\mathbf{M}, \mathbf{B}) \in \mathbb{F}_2^q$. The security decoder recovers the message from the output of the reliability decoder via $\widehat{\mathbf{M}} = \psi_s(\widehat{\mathbf{V}})$. Moreover, the reliability encoder e_r generates the transmitted symbol X_i using the output of the security encoder $\mathbf{V}$ and past feedback from Bob and Eve as

$$X_i = e_r(\mathbf{V}, \tilde{Y}_1, \ldots, \tilde{Y}_{i-1}, \tilde{Z}_1, \ldots, \tilde{Z}_{i-1}), \quad \forall i \in [1\!:\!1\!:\!n]. \tag{2}$$

After n channel uses, the reliability decoder d_r estimates $\mathbf{V}$ based on the received noisy codewords as

$$\widehat{\mathbf{V}} = d_r(Y_1, \ldots, Y_n). \tag{3}$$

Denote the effective code rate of the reliability layer as $R_r = q/n$. The functions e_r and d_r represent learned models.

Codes for the wiretap channel aim to ensure reliable transmission to Bob while maintaining the confidentiality of the message. To quantify information leakage, we define the security metric as $L_{\text{eve}} := I(\mathbf{M}; \mathbf{Z}^n)$, representing the mutual information (MI) between the message and the observations at the Eve, measured in bits. Moreover, define $I_{\text{bob}} := I(\mathbf{M}; \mathbf{Y}^n)$. For reliability, we consider the block error rate (BLER) at Bob, defined as $\mathrm{BLER} := \mathbb{P}(\mathbf{M} \neq \widehat{\mathbf{M}})$. The secrecy capacity is the maximum rate at which information can be transmitted both reliably and securely over a wiretap channel in the asymptotic regime, as $n \to \infty$. For the Gaussian wiretap channel without feedback, the secrecy capacity is given by [12]

$$C_s = \frac{1}{2}\left[\log\left(1 + P/\sigma^2_{Y,f}\right) - \log\left(1 + P/\sigma^2_{Z,f}\right)\right]^+. \tag{4}$$

Thus, the secrecy capacity is zero if $\sigma^2_{Y,f} \geq \sigma^2_{Z,f}$, as in a RD-WTC. We next provide the secrecy capacity with noiseless feedback.

Corollary 1 ([4]). *For a reversely-degraded wiretap channel with $\mathbb{P}(Y, Z|X) = \mathbb{P}(Z|X)\mathbb{P}(Y|Z)$, the secrecy capacity with noiseless feedback is*

$$C_s = \max_{\mathbb{P}_{(X)}} \min[H(Y|Z), I(X; Y)]. \tag{5}$$

Thus, with feedback, the secrecy capacity can become positive, unlike the non-feedback secrecy capacity in (4), which is zero in this regime. We demonstrate this feedback lunch property by constructing practical finite-length feedback codes that are effective in this RD-WTC-F.

III. Prior Work

In this section, we introduce two coding schemes for the AWGN with *noiseless* feedback: the Schalkwijk–Kailath (SK) scheme [13] and the POWERBLAST (PB) scheme [10]. For the WTC-F, we assume feedback only from Bob, where these schemes serve as the reliability layer (e_r, d_r) and become WTC-SK and WTC-PB once combined with the security layer.

The SK scheme asymptotically achieves the AWGN-F channel capacity [13] and the optimal Gaussian WTC-F secrecy capacity [14], i.e., $C_s = \frac{1}{2}\log\left(1 + \frac{P}{\sigma^2_{Y,f}}\right)$. In the first round, the message bits $\mathbf{M} \in \mathbb{F}_2^k$ (corresponding to $\mathbf{V} \in \mathbb{F}_2^q$ in WTC-SK) are mapped to a pulse amplitude modulation (PAM) symbol $\Theta \in \{\pm 1\eta, \pm 3\eta, \ldots, \pm(2^k - 1)\eta\}$, where $\eta = \sqrt{\frac{3}{2^{2k}-1}}$ ensures unit power. The transmitted signal is scaled to $X_1 = \sqrt{P}\Theta$. In the subsequent rounds, the transmitter iteratively sends the scaled estimation error $X_i = \sqrt{\frac{P}{D_{i-1}}}\epsilon_{i-1} = \sqrt{\frac{P}{D_{i-1}}}\left(\widehat{\Theta}_{i-1} - \Theta\right)$, where $\widehat{\Theta}_i$ is the estimated symbol at time i. The estimation error is $\epsilon_i = \widehat{\Theta}_i - \Theta$ with mean squared error $D_i = \mathbb{E}(\epsilon_i^2)$. At the receiver, after the first round, the linear minimum mean square error estimate is $\widehat{\Theta}_1 = \frac{\sqrt{P}}{P + \sigma^2_{Y,f}} Y_1$. In the subsequent rounds, the receiver estimates the error as $\hat{\epsilon}_{i-1} = \frac{\sqrt{PD_{i-1}}}{P + \sigma^2_{Y,f}} Y_i$ and updates $\widehat{\Theta}_i = \widehat{\Theta}_{i-1} - \hat{\epsilon}_{i-1}$. Finally, the receiver decodes by mapping $\widehat{\Theta}_n$ to the nearest PAM symbol. The SK scheme achieves doubly exponential error decay in n in terms of BLER.

The PB [10] scheme is a variation of the SK that differs only in the final round, where the transmitter sends a discrete symbol for the estimated PAM index error $X_n = \sqrt{\frac{P}{D_I}}(I(\widehat{\Theta}_{n-1}) - I(\Theta))$ with $D_I = \mathbb{E}[(I(\widehat{\Theta}_{n-1}) - I(\Theta))^2]$ and $I(\Theta) \in [1\!:\!1\!:\!2^k]$. At the receiver, a maximum a posteriori decoder is used to recover the PAM symbol.

IV. Proposed Coding Schemes

We next propose practical feedback coding schemes for wiretap channels with feedback by extending the modular coding scheme. To measure their security performance, we also estimate the information leakage using neural networks, which later is used to improve their overall performance.

A. Security Layer (φ_s, ψ_s)

To limit the information leakage, we use 2-universal hash functions (2-UHF) φ_s and its inverse ψ_s, as defined below.

Definition 1 (2-UHF [15]). *Let $\mathcal{X}$ and $\mathcal{Y}$ be finite sets. A family of functions $\mathcal{H}$, where each $H : \mathcal{X} \mapsto \mathcal{Y}$, is 2-universal if $\forall x_1, x_2 \in \mathcal{X}$ with $x_1 \neq x_2$, we have*

$$\mathbb{P}_{H\sim\mathcal{H}}(H(x_1) = H(x_2)) \leq \mathit{card}(\mathcal{Y})^{-1} \tag{6}$$

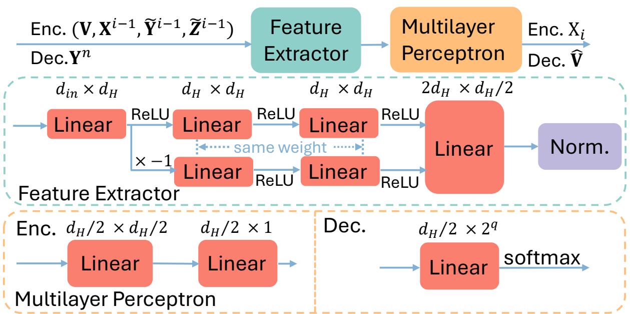

Fig. 2. Detailed structure of the reliability layer.

where H is chosen uniformly at random from $\mathcal{H}$.

Let the seed space be $\mathcal{S} := \mathbb{F}_2^q \backslash \{\mathbf{0}\}$, where the seed $\mathbf{S} \in \mathcal{S}$ is shared among the legitimate parties. For each message $\mathbf{M} \in \mathbb{F}_2^k$, Alice samples a uniformly distributed random bit sequence $\mathbf{B} \in \mathbb{F}_2^{q-k}$. The security encoder φ_s is defined as

$$\varphi_s : (\mathbf{M}, \mathbf{B}) \mapsto \mathbf{S}^{-1} \odot (\mathbf{M} \| \mathbf{B}) \tag{7}$$

where $(\cdot \| \cdot)$ denotes the concatenation of two sequences, and $\odot$ represents multiplication in $\mathrm{GF}(2^q)$. The corresponding security decoder recovers the estimated message $\widehat{\mathbf{M}}$ using

$$\psi_s : \widehat{\mathbf{V}} \mapsto (\mathbf{S} \odot \widehat{\mathbf{V}})_k \tag{8}$$

where $(\cdot)_k$ denotes selection of the k most significant bits. The security layer (φ_s, ψ_s) is used in all WTC-based codes.

B. Reliability Layer (e_r, d_r)

The reliability layer (e_r, d_r) can be implemented using neural networks. Unlike the original Lightcode [10], WTC-Lightcode incorporates feedback from both Bob and Eve, as in the broadcast case [16] (Fig. 1), and augments Lightcode with the security layer (φ_s, ψ_s). Both e_r and d_r consist of a feature extractor and a multi-layer perceptron, as shown in Fig. 2, where $d_H = 32$ and $d_{\text{in}} = k + 3(n-1)$ with unused positions zero-padded. As the blocklength n increases, the parameter size grows rapidly and makes training difficult, so we focus on short blocklengths, while longer ones can be divided into smaller chunks and transmitted separately [10]. The reliability encoder and decoder are trained jointly to minimize the categorical cross entropy (CCE) $J_{\text{CCE}} = -\frac{1}{B}\sum_{i=1}^{B}\left(\sum_{j=1}^{C_V} p_{ij}\log(\hat{p}_{ij})\right)$, where B is the batch size, $C_V = 2^q$, $p_{ij} = 1$ if class j is correct for instance i and 0 otherwise, and $\hat{p}_{ij} \in \mathbb{R}$ is the predicted probability.

C. Information Leakage

The joint or marginal distributions are often unknown in closed form, making direct computation of MI infeasible. We therefore approximate I_{bob} using Mutual Information Neural Estimator (MINE) [17] and L_{eve} with f-divergence discriminative mutual information estimators (f-DIME) [18].

MINE uses the dual representation of the Kullback-Leibler (KL) divergence, yielding the lower bound

$$\hat{I}_{\text{bob}} := \mathbb{E}_{\mathbb{P}_{(\mathbf{M},\mathbf{Y}^n)}}[T(\mathbf{m},\mathbf{y}^n)] - \log\left(\mathbb{E}_{\mathbb{P}_{\mathbf{M}}\mathbb{P}_{\mathbf{Y}^n}}[e^{T(\mathbf{m},\mathbf{y}^n)}]\right) \tag{9}$$

where $T : \mathbb{F}_2^k \times \mathbb{R}^n \to \mathbb{R}$ is a neural network-parameterized score function. We use MINE for short blocklengths due to its high variance in high dimensions, and validate the results with other lower-bound estimators such as NWJ and SMILE [17], [19]. In contrast, f-DIME extends the variational framework to general f-divergences and uses derangement sampling, yielding more accurate and lower-variance MI estimates via

$$\hat{T} = \arg\max_{T} \mathbb{E}_{\mathbb{P}_{(\mathbf{M},\mathbf{Z}^n)}}\left[T(\mathbf{m},\mathbf{z}^n) - f^*\left(T(\mathbf{m},\varsigma(\mathbf{z}^n))\right)\right], \tag{10}$$

$$\hat{L}_{\text{eve}} := \mathbb{E}_{\mathbb{P}_{(\mathbf{M},\mathbf{Z}^n)}}\left[\log\left((f^*)'(\hat{T}(\mathbf{m},\mathbf{z}^n))\right)\right] \tag{11}$$

where f^* is the Fenchel conjugate of $f : \mathbb{R}_+ \to \mathbb{R}$ with $f(1) = 0$, $(f^*)'$ its derivative, ς a permutation function such that $\mathbb{P}_{\varsigma(\mathbf{Z}^n)}(\varsigma(\mathbf{z}^n)|\mathbf{m}) = \mathbb{P}_{\mathbf{Z}^n}(\mathbf{z}^n)$, and $T : \mathbb{F}_2^k \times \mathbb{R}^n \to \mathbb{R}$ a neural network. We take the maximum among f-DIME variants based on the KL divergence, Hellinger distance squared, and generative adversarial networks (GAN) divergence[1].

D. Trade-off Loss

We next incorporate the information leakage $\hat{L}_{\text{eve}}$ into the loss function by formulating a constrained optimization problem as

$$\min_{(e_r,d_r)} \; J_{\text{CCE}} \quad \text{s.t. } \hat{L}_{\text{eve}} \le \tau \tag{12}$$

where τ is the target leakage threshold. To address this, we introduce the trade-off loss as

$$J_\beta = J_{\text{CCE}} + \beta(\hat{L}_{\text{eve}} - \tau) \tag{13}$$

with β updated as $\beta \leftarrow [\beta + \eta(\hat{L}_{\text{eve}} - \tau)]^+$, starting from $\beta = 0$ and learning rate $\eta = 0.01$. The training consists of two interleaved phases. We initialize the model (e_r, d_r) with weights pre-trained using only the CCE loss. In Phase 1, the f-DIME is retrained every 50 batches to estimate the current information leakage. In Phase 2, (e_r, d_r) is updated using the trade-off loss J_β.

V. Reversely-Degraded Feedback Wiretap Code Simulation Results

We next evaluate the seeded modular learned feedback codes in terms of security (information leakage) and reliability (BLER). The encoder and decoder are jointly trained to minimize the CCE loss, and we further study the trade-off between reliability and information leakage by incorporating leakage into the loss. We observe that leakage depends on the random seed, so we fix s as shared randomness between legitimate parties. During training, Bob and Eve share the same forward SNR, while during testing, Bob's SNR is fixed and Eve's SNR is varied[2].

A. Secrecy Rate Effects

We consider the baseline setting with noiseless feedback, CCE loss, and message length $k = 3$. The random seeds

[1]CLUB [20] was also tested as an upper bound estimator but proved inaccurate, often exceeding the entropy $H(\mathbf{M}) = 3$.

[2]Simulations are implemented in PyTorch on an NVIDIA Tesla V100 GPU. Code and parameters: https://github.com/zyy-cc/wiretap-feedback.git

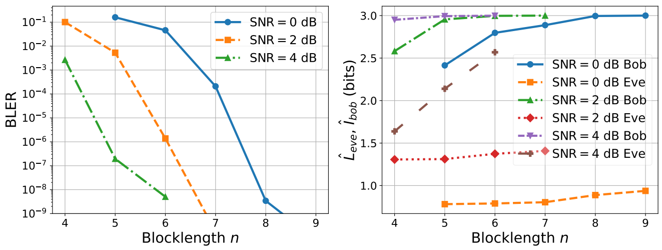

Fig. 3. BLER (left) and estimated $\hat{I}_{\text{bob}}$, $\hat{L}_{\text{eve}}$ (right) versus blocklength under SNR $= S_{Y,f} = S_{Z,f}$ with noiseless feedback.

$\mathbf{s} = [1,0,1]$, $[1,1,0,1]$, and $[1,1,0,1,0]$ are fixed for $q = 3,4,5$ and shared between the legitimate parties. As shown in Fig. 3, we first study the effect of blocklength with $q = 3$ (no random bits $\mathbf{B}$) and equal SNR ($S_{Y,f} = S_{Z,f}$) on reliability and security in WTC-Lightcode. Each additional round of error correction improves reliability, with BLER decreasing quickly and $\hat{I}_{\text{bob}}$ approaching $H(\mathbf{M}) = 3$ bits as Bob's observations nearly reveal the complete message. In contrast, the information leakage remains relatively small at low SNRs (0 and 2 dB) and increases only slightly with the blocklength n. At higher SNR (4 dB), leakage grows with blocklength, as Eve's reliable observations also influence the encoder output. This effect can be mitigated by adding random bits or imposing a leakage constraint. Notably, when Bob and Eve experience the same forward SNR, $\hat{I}_{\text{bob}}$ typically exceeds $\hat{L}_{\text{eve}}$, even in the RD case. This suggests that feedback creates additional shared randomness between Alice and Bob, providing feedback lunch, as defined above. We measure this effect by the *security-advantage gain (SAG)* defined below.

Definition 2 (Security-advantage Gain)**.** *Given I_{bob} at forward SNR $S_{Y,f}$, let $S_{Z,f}$ be the smallest SNR for Eve such that the leakage satisfies $|I_{bob}(S_{Y,f}) - L_{eve}(S_{Z,f})| < \varepsilon$ for a small threshold $\varepsilon > 0$. The SAG is defined as*

$$\Lambda_{(k,q,n,\mathrm{S}_{Y,f})} := \frac{1}{2}\log_2 \frac{1+S_{Z,f}}{1+S_{Y,f}}. \tag{14}$$

SAG roughly quantifies Eve's extra effort to match Bob's information level. Beyond this threshold, feedback offers no advantage, making secure communication impossible.

Remark 1. The additional effort required for Eve to match the information level at Bob for feedback wiretap channels actually depends on the code rate, forward SNRs, feedback SNRs, and the length of the random sequence $q-k$ because of the finite blocklengths considered. However, SAG serves as a practical measure of the feedback lunch, which is more accurate than existing practical metrics in the literature, such as the metric that measures the SNR differences. Thus, we consider SAG for our analyses.

We observe that the baseline SAG values $\Lambda_{(3,3,9,0)} = 0.89$, $\Lambda_{(3,3,7,2)} = 1.41$, and $\Lambda_{(3,3,6,4)} = 0.67$ all arise from channel output feedback. Next, we study how the random bit length $q-k$ affects performance, with $\hat{I}_{\text{bob}}$ and $\hat{L}_{\text{eve}}$ evaluated at $\mathrm{S}_{Y,f}$ (Table I). For fixed n, adding random bits reduces leakage and yields higher SAG than the no-random-bit case $q = k$. However, longer random sequences do not always help, as maximizing SAG requires both low BLER (or higher I_{bob}) and low leakage (L_{eve}). Thus, selecting $q-k$ to maximize SAG at a given SNR is important to improve overall performance. For instance, we observe that $\Lambda_{(3,4,9,0)} = 4.48$ exceeds $\Lambda_{(3,5,9,0)} = 3.99$.

TABLE I
EFFECTS OF RANDOM BIT SEQUENCE WITH $k = 3$.

$\mathrm{S}_{Y,f}$ (dB)	q	n	BLER	$\hat{I}_{\text{bob}}$	$\hat{L}_{\text{eve}}$	Λ
0	4	9	1.35×10^{-4}	2.97	0.34	4.48
0	5	9	0.16	2.23	0.14	3.99
2	4	7	6.91×10^{-4}	2.97	1.14	4.96
2	5	7	0.11	2.33	0.43	4.13
4	4	6	5.25×10^{-5}	2.99	1.69	2.75
4	5	6	2.24×10^{-2}	2.64	0.80	2.42

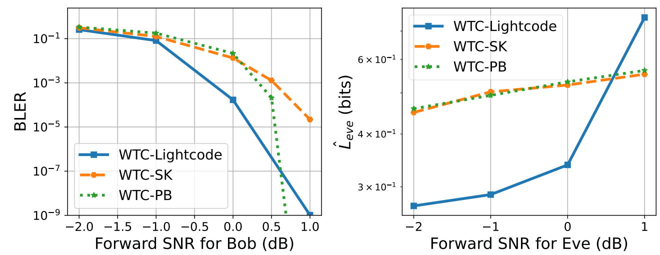

Fig. 4. BLER versus forward SNR at Bob (left) and $\hat{L}_{\text{eve}}$ versus forward SNR at Eve (right). Each data point is obtained under the condition $\mathrm{S}_{Y,f} = \mathrm{S}_{Z,f}$.

B. Comparisons with Existing Feedback Codes

We next compare the proposed WTC-Lightcode with other feedback codes, such as WTC-SK and WTC-PB that also exhibit a free lunch property. For example, we observe $\Lambda_{(3,4,9,0)} = 4.15$ for WTC-SK, slightly below WTC-Lightcode (4.48). We set $q - k = 1$ with message length $k = 3$ and blocklength $n = 9$, which leverages the randomness introduced by the security layer while maintaining high BLER performance. We depict reliability and information leakage performance in Fig. 4. We observe that WTC-Lightcode performs better at low SNRs in both BLER and information leakage, while WTC-PB outperforms WTC-Lightcode in BLER at high SNRs, achieves lower leakage than WTC-Lightcode, and is thus the preferred choice in that regime. Moreover, WTC-SK exhibits worse BLER performance than WTC-Lightcode, while achieving similar information leakage to WTC-PB. This follows because WTC-SK and WTC-PB are identical during the first $n-1$ rounds, and in the final round, both transmit only the estimated error from previous rounds instead of the message $\mathbf{M}$. In contrast, WTC-Lightcode is a nonlinear scheme that incorporates feedback from Eve, resulting in different information leakage.

C. Noisy Feedback Effects

The analytical linear codes WTC-SK and WTC-PB with noisy feedback fail to achieve a positive secure communication rate, motivating the nonlinear learned WTC-Lightcode, which is more robust to feedback noise [21]. Table II considers a message length of $k = 3$, with $\hat{L}_{\text{eve}}$ evaluated under equal forward ($\mathrm{S}_{Y,f} = \mathrm{S}_{Z,f}$) and feedback ($\mathrm{S}_{Y,fb} = \mathrm{S}_{Z,fb}$) SNRs.

TABLE II
RELIABILITY AND SECURITY WITH NOISY FEEDBACK WITH $k = 3$.

$\mathrm{S}_{Y,f}$	$\mathrm{S}_{Y,fb}$	q	n	BLER	$\hat{I}_{\text{bob}}$	$\hat{L}_{\text{eve}}$	Λ
0	20	3	8	2.21×10^{-3}	2.97	1.34	2.50
0	10	3	8	1.82×10^{-2}	2.91	2.10	0.53
0	5	3	8	3.8×10^{-2}	2.83	2.32	0.29
4	20	3	6	4.02×10^{-6}	2.99	2.56	0.82
4	10	3	6	8.44×10^{-4}	2.99	2.81	0.53
4	5	3	6	4.16×10^{-3}	2.98	2.88	0.12
1	20	4	9	3×10^{-3}	2.96	1.07	3.73
1	10	4	9	2.8×10^{-2}	2.82	2.17	0.44
1	5	4	9	4.1×10^{-2}	2.78	2.51	0.09

We test both low and high SNR regimes, with and without random bits, where WTC-Lightcode maintains good BLER performance. We observe that noisier feedback channels increasingly disrupt the shared randomness between legitimate parties, leading to greater information leakage. Nevertheless, the legitimate parties still achieve a positive SAG even under noisy feedback.

D. Security-Reliability Trade-off Loss Function Effects

We observe that information leakage in WTC-Lightcode increases with forward SNRs and feedback noise. To mitigate this, we introduce the trade-off loss that constrains leakage while preserving reliability. By specifying a target leakage τ in (13), the model is trained to approach this value. As shown in Fig. 5, for $k = 3$, we vary τ to reduce leakage under different SNRs. For $\mathrm{S}_{Y,f} = \mathrm{S}_{Z,f} = 1$ dB with noiseless feedback (FB), $\tau \in [0.1 : 0.1 : 0.6]$ (original leakage 0.76 bits). With noisy feedback $\mathrm{S}_{Y,fb} = \mathrm{S}_{Z,fb} = 5$ dB, $\tau \in [0.9 : 0.3 : 2.4]$ (original leakage 2.51 bits). For $\mathrm{S}_{Y,f} = \mathrm{S}_{Z,f} = 4$ dB, $\tau \in [1 : 0.2 : 2.4]$ (original leakage 2.57 bits). The main trend is that reducing leakage increases BLER. By tuning τ, information leakage can be controlled even with noisy feedback, ensuring secure communication while maintaining low BLER.

VI. CONCLUSION

We studied seeded modular code designs for the Gaussian wiretap channel with feedback, using UHF for the security layer and WTC-Lightcode for the reliability layer. With (noisy) feedback, even in the challenging reversely-degraded case, legitimate parties have a SAG over Eve thanks to the feedback lunch. We also illustrated the effects of different parameters and compared the performance of different feedback code designs. Furthermore, we incorporated information leakage into the training loss to enable a flexible and better trade-off between BLER performance and information leakage. Our results motivate new feedback code designs for secure ISAC applications considered for sixth-generation (6G) systems, where channel feedback is intrinsic. In the future, we will adapt hybrid code constructions in [22] to enable larger blocklengths for the deep-learned feedback codes.

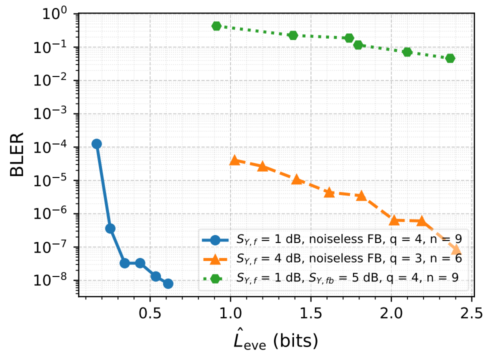

Fig. 5. Trade-off between the BLER and information leakage.

REFERENCES

[1] M. Bloch and J. Barros, *Physical-Layer Security: From Information Theory to Security Engineering*. Cambridge University Press, 2011.
[2] M. Bloch *et al.*, "An overview of information-theoretic security and privacy: Metrics, limits and applications," *IEEE J. Sel. Areas Inf. Theory (JSAC)*, vol. 2, no. 1, pp. 5–22, 2021.
[3] A. D. Wyner, "The wire-tap channel," *Bell Sys. Techn. J.*, vol. 54, no. 8, pp. 1355–1387, 1975.
[4] R. Ahlswede and N. Cai, "Transmission, identification and common randomness capacities for wire-tape channels with secure feedback from the decoder," in *General Theory of Information Transfer and Combinatorics*. Springer, 2006, pp. 258–275.
[5] O. Günlü, M. Bloch, R. F. Schaefer, and A. Yener, "Secure integrated sensing and communication," *IEEE J. Selected in Inf. Theory (JSAIT)*, vol. 4, pp. 40–53, 2023.
[6] M. Bellare, S. Tessaro, and A. Vardy, "Semantic security for the wiretap channel," in *Adv. in Crypt. (CRYPTO)*, R. Safavi-Naini and R. Canetti, Eds. Springer Berlin Heidelberg, 2012, pp. 294–311.
[7] V. Rana and R. A. Chou, "Short blocklength wiretap channel codes via deep learning: Design and performance evaluation," *IEEE Trans. on Commun. (TCOM)*, vol. 71, no. 3, pp. 1462–1474, 2023.
[8] D. Seifert, O. Günlü, and R. F. Schaefer, "Deep learning-based codes for wiretap fading channels," in *IEEE Int. Symp. on Personal, Indoor Mobile Radio Commun. (PIMRC)*, 2025.
[9] G. Fettweis *et al.*, "Joint communications & sensing – Common radio-communications and sensor technology," *VDE Positionspapier*, 2021.
[10] S. K. Ankireddy, K. Narayanan, and H. Kim, "LIGHTCODE: Light analytical and neural codes for channels with feedback," *IEEE J. Sel. Areas Commun. (JSAC)*, pp. 1–1, 2025.
[11] X. Zhang, H. Gu, L. Fan, K. Chen, and Q. Yang, "No free lunch theorem for security and utility in federated learning," *ACM Trans. on Intell. Syst. and Techno. (TIST)*, vol. 14, no. 1, pp. 1–35, 2022.
[12] S. K. Leung-Yan-Cheong, *Multi-User and Wiretape Channels Including Feedback*. Stanford University, 1976.
[13] J. Schalkwijk and T. Kailath, "A coding scheme for additive noise channels with feedback–i: No bandwidth constraint," *IEEE Trans. on Inf. Theory (T-IT)*, vol. 12, no. 2, pp. 172–182, 1966.
[14] D. Gunduz, D. R. Brown, and H. V. Poor, "Secret communication with feedback," in *Int. Symp. Inf. Theory Appl. (ISITA)*, 2008, pp. 1–6.
[15] J. L. Carter and M. N. Wegman, "Universal classes of hash functions," in *ACM Symp. on Theory of Comput. (STOC)*, 1977, pp. 106–112.
[16] Y. Zhou and N. Devroye, "Learned codes for broadcast channels with feedback," in *ICC 2025-IEEE International Conference on Communications*. IEEE, 2025, pp. 3803–3808.
[17] M. Belghazi *et al.*, "Mutual information neural estimation," in *PMLR Int. Conf. on Machine Learning (ICML)*, 2018, pp. 531–540.
[18] N. A. Letizia, N. Novello, and A. M. Tonello, "f-DIME: Mutual information estimation via f-divergence and data derangements," *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 37, pp. 105 114–105 150, 2024.
[19] J. Song and S. Ermon, "Understanding the limitations of variational mutual information estimators," in *International Conference on Learning Representations*, 2020.
[20] P. Cheng, W. Hao, S. Dai, J. Liu, Z. Gan, and L. Carin, "CLUB: A contrastive log-ratio upper bound of mutual information," in *Int. Conf. Mach. Learn. (ICML)*, 2020, pp. 1779–1788.
[21] Y.-H. Kim, A. Lapidoth, and T. Weissman, "The Gaussian channel with noisy feedback," in *IEEE Int. Symp. Inf. Theory (ISIT)*, 2007, pp. 1416–1420.
[22] O. Günlü, R. Fritschek, and R. F. Schaefer, "Concatenated classic and neural (CCN) codes: ConcatenatedAE," in *IEEE Wireless Commun. Netw. Conf. (WCNC)*, 2023, pp. 1–6.